

From Flood Maps to Action: An Open-Data Audit of Flood Preparedness and Early-Warning Evidence in Dire Dawa, Ethiopia

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Abstract

Dire Dawa faces recurrent flood risk, while the public record contains no single, current account of whether warnings translate into safe action across the city. This report audits publicly available evidence on flood preparedness and early warning in Dire Dawa. The study uses a systematic source inventory, finding-level extraction matrix, and transparent quality screening across seven preparedness stages: risk monitoring and hazard information; forecasting and warning generation; warning dissemination; warning receipt, understanding, and accessibility; protective action and evacuation; shelter, emergency health, and service continuity; and recovery, livelihoods, and learning. The evidence documents historical community-based warning initiatives, a city-level preparedness diagnostic, household preparedness research, local qualitative findings, a technical forecasting approach, and vulnerability-related household data. However, the sources are fragmented, differently scoped, and often historical. They are insufficient to verify current operational coverage, warning comprehension, evacuation readiness, shelter functionality, emergency-health continuity, or end-to-end warning performance. The central conclusion is that public evidence supports targeted field validation rather than assumptions about either system failure or effectiveness. Immediate no-regret actions can improve evidence coordination, validation tools, message design, and documentation. Decisions about alert-channel expansion, evacuation routes, shelters, kebele targeting, or coverage claims require current, location-specific validation. The report therefore treats missing public evidence as an evidence gap, not as proof that a service or capability is absent.

Keywords

Dire Dawa; flood preparedness; early warning systems; disaster risk reduction; open data; urban resilience; evidence audit.

1. Introduction

Flooding matters in Dire Dawa because the city and surrounding communities face recurring exposure to intense rainfall, drainage pressures, and rapid flood impacts. Public studies, city diagnostics, community projects, household research, and technical assessments have all treated flooding as a meaningful risk. Yet these sources do not form a single, current operational record. They describe different places, populations, periods, and components of preparedness.

A hazard map can show where water may go, but it cannot by itself show whether an alert is generated in time, whether it reaches households, whether people understand it, whether they can move safely, or whether shelter and health services can function during the event. Preparedness is therefore a chain. A weakness or uncertainty at any link can limit the practical value of the other links. A siren, forecast model, or map should not be treated as a complete warning system without evidence about decision authority, communication, accessibility, protective action, and recovery.

The central evidence problem is fragmentation. One source may document a historical community program; another may assess city-level capacity; another may measure household perceptions; another may propose a technical model. These sources are valuable, but they cannot be combined without distinguishing what each actually demonstrates. The public record also contains different levels of geographic specificity, from named kebeles and communities to city-level or comparative datasets. This makes it necessary to assess not only what a source says, but also where, when, and how it supports a conclusion.

This report asks: What does publicly available evidence show about flood preparedness and early-warning systems in Dire Dawa, what cannot be verified from available evidence, and what information is needed for better local action?

The four objectives are to: (1) identify and organize publicly accessible evidence related to flood preparedness and early warning in Dire Dawa; (2) assess evidence across the full preparedness chain; (3) evaluate recency, geographic specificity, accessibility, methodological strength, actionability, and reusability; and (4) identify decision-critical evidence gaps requiring responsible field validation before interventions are designed or targeted.

This study does not collect primary data, verify current system performance, rank kebeles or communities, identify evacuation routes or shelter capacity, expose sensitive infrastructure or personal information, or constitute a government risk assessment or operational flood model. It is a secondary-evidence audit. Where current, location-specific evidence is unavailable, the report does not infer that a service is absent. It records the limitation as an evidence gap.

2. Methods

2.1 Study design

The study uses a structured secondary-data and open-evidence audit. The unit of analysis is the finding, not merely the report. Sources were inventoried, relevant findings were extracted, and each finding was mapped to one or more stages in the seven-domain preparedness chain. The design is intended to be reproducible: another reviewer can inspect the source inventory, trace a finding to its source ID, review its limitations, and see how it informed the analysis memo and report.

2.2 Source identification

Source identification covered publicly accessible academic literature, institutional and government material, city diagnostics, technical studies, project and case-study documents, datasets or dataset documentation, and publicly described planning or monitoring records. The search logic was guided by combinations of Dire Dawa, flood, flash flood, preparedness, early warning, forecasting, evacuation, shelter, emergency health, vulnerability, resilience, drainage, and recovery. The inventory records the source type, publication year, evidence period, geography, method, accessibility, and decision taken.

2.3 Eligibility criteria

Sources were included when they contained direct Dire Dawa evidence or usable evidence about a documented local activity, local population, local technical analysis, or relevant city-level preparedness condition. Sources were retained as contextual or potential evidence when their local relevance required verification. General references without direct findings were excluded from primary synthesis. Missing administrative, monitoring, shelter, health, evacuation, and coordination records were listed as evidence needs rather than treated as negative findings.

The study uses only public or lawfully accessible secondary sources. It excludes primary household or institutional data collection, unverified claims of current operational performance, rankings of localities, and sensitive location information that could expose vulnerable people or critical infrastructure.

2.4 Evidence extraction

The source inventory records one row per source. The preparedness evidence matrix records one row per finding and includes source ID, domain, subdomain, objective finding text, evidence year, publication year, geography, population or system, evidence type, evidence status, confidence, actionability, caveat, page or section reference, and an accurate paraphrase. Evidence status uses five labels: Documented, Partial, Historical, Unverified current status, and Evidence gap.

2.5 Evidence-quality screening

Each included source was screened on a 1-to-5 scale for recency, geographic specificity, method strength, public accessibility, actionability, and reusability. The average is the sum of the six criteria divided by six. Scores of 4.0-5.0 are classified as strong operational evidence; 3.0-3.9 as useful but requiring qualification; 2.0-2.9 as contextual or historical; and below 2.0 as background only. This is a transparent internal screening framework developed for this evidence audit, not an internationally validated scale. Unlisted recency periods were treated conservatively through interpolation, as documented in Appendix C.

2.6 Analysis approach

The analysis uses thematic synthesis along the seven-domain warning chain. For each domain, the report states what is documented, the source year and geography, the strength and status of the evidence, and what remains unknown. The analysis memo then distinguishes safe no-regret actions from decisions that require current field validation. This prevents

a technical model from being treated as an operational service, a past project from being treated as a current system, or missing evidence from being treated as proof of absence.

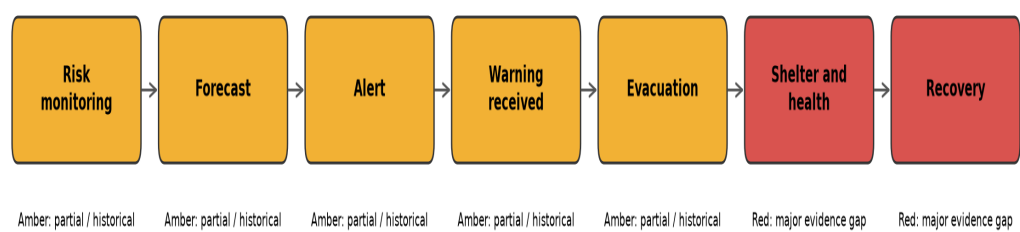
2.7 Ethics and limitations

The study uses secondary public sources and does not collect personal data. It does not draw conclusions about current readiness of particular communities where current, location-specific evidence is unavailable. Limitations include dependence on public sources, possible missing official documents, historical studies, heterogeneous methods, restricted datasets, uncertain current status, and no primary validation. These limitations are part of the result: they define what can responsibly be claimed and what must be checked before intervention design.

3. Results

Figures reproduced from the verified Step 6 visual package.

Figure 1. Dire Dawa preparedness chain and evidence status



Source: 03_preparedness_evidence_matrix.csv; status reflects the strongest usable evidence, not current system performance.

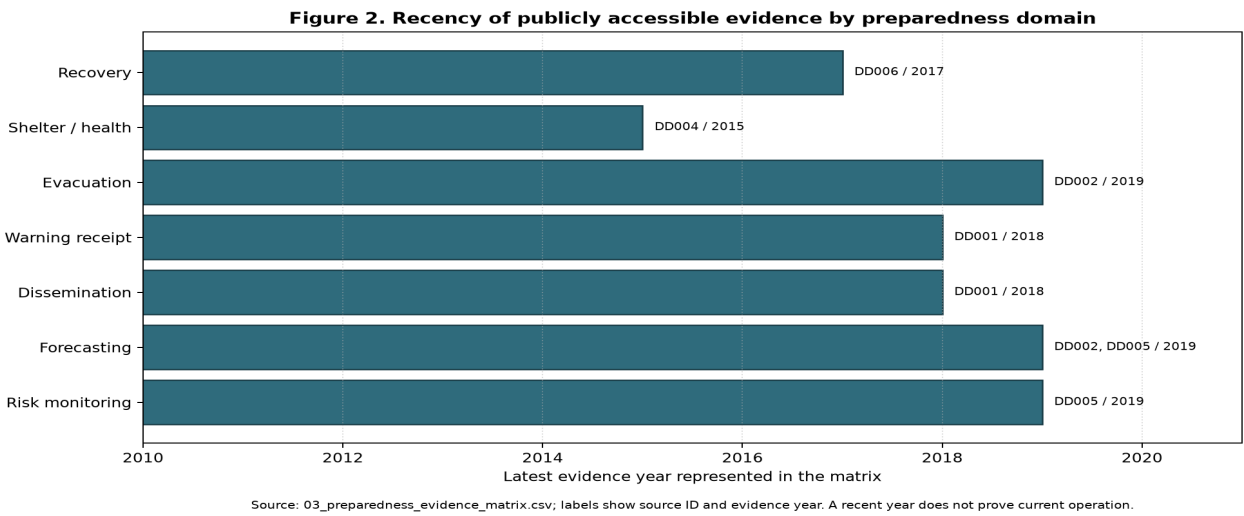
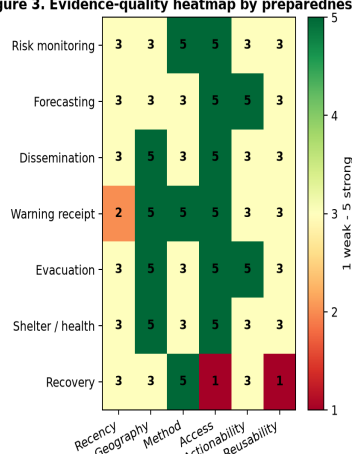


Figure 3. Evidence-quality heatmap by preparedness domain



Highest available score among included source(s) mapped to each domain

Source: 04_evidence_quality_scorecard.xlsx; domain values are screening summaries, not system-level scores. Source IDs: Risk monitoring (DD005/DD006); Forecasting (DD002/DD005); Dissemination (DD001/DD004); Warning receipt (DD001/DD003); Evacuation (DD002/DD004); Shelter / health (DD004); Recovery (DD006)

Figure 4. Research-to-action pathway

Evidence finding	What it means	Safe immediate action	Field validation needed
Historical warning methods	Community-based approaches are plausible, but not verified.	Prepare a standardized audit tool.	Test functionality, maintenance, reach, accessibility, and lead time today.
Weak route and shelter evidence	Evacuation cannot be planned safely from public information alone.	Identify data owners and coordinate verification.	Map and inspect routes, assembly points, shelters, capacity, and access.
Old household readiness evidence	Preparedness barriers and communication needs are not current.	Review and update communication materials.	Conduct an inclusive household and community assessment.

Source: 05_analysis_memo.md and 03_preparedness_evidence_matrix.csv; actions are deliberately separated from field-dependent decisions.

The results are presented in the agreed preparedness-chain order. The descriptions below report the evidence base before drawing broader implications in the Discussion.

3.1 Risk monitoring and hazard information

The inventory contains household-level evidence relevant to flood exposure, poverty, housing conditions, and coping capacity in urban Ethiopia, including Dire Dawa-related data (World Bank, 2017). A Dire Dawa technical study developed a flood-hazard forecasting approach using the Soil Conservation Service-Curve Number (SCS-CN) method (Abebe, 2017). These sources document risk information and analytical work, but they do not constitute a current, integrated city monitoring system.

The household evidence dates from 2017 and covers Addis Ababa and Dire Dawa. The technical study is focused on Dire Dawa and dates from 2019, but it presents a modelling approach rather than a verified network of monitoring stations, current thresholds, or routine public information products. Evidence strength is therefore mixed: the existence of risk analysis is documented, while present monitoring coverage and data integration are not.

Unknowns include current rainfall and water-level records, data-sharing arrangements, flood-prone-area mapping, exposure updates, trigger thresholds, and how monitoring information is converted into preparedness decisions. The matrix classifies current operational monitoring records and usable location-specific exposure maps as evidence gaps.

3.2 Forecasting and warning generation

The technical study provides local evidence of a flood-hazard forecasting and early-warning approach (Abebe, 2017). The World Bank city diagnostic reported that city-level early-warning capacity needed strengthening and that warnings could arrive too late for residents to reach safety (World Bank, 2019). Together, these sources document technical and strategic attention to forecasting and warning generation.

The technical evidence is city-focused and recent relative to other sources, but it does not verify forecast accuracy, institutional ownership, trigger governance, or public alerting. The diagnostic is an official city-level assessment, but it is not a current event-by-event performance audit. Evidence strength is therefore partial for operational planning and

stronger for identifying the need for validation.

The public record does not verify who currently authorizes alerts, which thresholds are used, what lead time is achieved, how warnings are logged, or whether backup procedures have been tested.

3.3 Warning dissemination

The Red Cross and Partners for Resilience case study documents community-managed warning arrangements using mobile phones, sirens, megaphones, local indicators, community committees, and community information centres (Climate Centre et al., 2010). A qualitative study also documented warning channels and mitigation measures in Kebeles 05 and 06 (Ejeta, 2018).

The case study describes a program and communities during a historical period extending through 2018, although it was published in 2010. The qualitative study was published in 2015 and is geographically limited to named kebeles. This is strong evidence that locally adapted dissemination approaches existed, but it is historical and not a current citywide coverage audit.

The public record does not verify current equipment condition, maintenance, responsible actors, backup power or communication, message languages, disability accessibility, household reach, or warning lead time.

3.4 Warning receipt, comprehension, and accessibility

The community case study documents multiple channels that could support local warning receipt, while a household survey found that preparedness was associated with perceived threat, perceived benefits and barriers, and cues to action (Climate Centre et al., 2010; Ejeta et al., 2016). These findings document the relevance of communication and behavioural conditions to preparedness.

The warning-channel evidence concerns specified communities and the survey concerns Dire Dawa households in 2015. The evidence is useful for understanding possible communication pathways and behavioural influences, but it does not test a current alert message or measure citywide comprehension. Its strength is therefore partial and historical for operational accessibility.

The current public record does not verify whether people with disabilities, older people, people with limited phone access, informal settlers, women, or livelihood-constrained households receive, understand, trust, and can act on warnings in time.

3.5 Protective action and evacuation

The city diagnostic reported concern that warnings could arrive too late for residents to reach safety (World Bank, 2019). The qualitative study identified gaps related to evacuation planning in Kebeles 05 and 06, while the household survey identified behavioural factors associated with preparedness (Ejeta, 2018; Ejeta et al., 2016).

The diagnostic provides city-level evidence from a defined assessment period through 2019; the qualitative findings are local to named kebeles and historical. The evidence documents concern and planning gaps, but not current route condition, travel times, transport capacity, assembly points, drills, or actual evacuation performance.

Current safe routes, accessible transport, assistance for people who cannot self-evacuate, and links between warning levels and protective actions remain unverified. No public citywide route or accessibility dataset was identified.

3.6 Shelter, emergency health, and service continuity

The qualitative study identified gaps in emergency health links, first-aid support, and temporary shelters in Kebeles 05 and 06 (Ejeta, 2018). This is the clearest direct local evidence in the reviewed sources for shelter and emergency-health limitations.

The evidence is a historical, geographically limited qualitative baseline. It does not provide a current citywide inventory of shelters, capacities, accessibility, supplies, staffing, referral links, or continuity arrangements. Evidence strength is therefore partial for identifying issues and weak for current operational conclusions.

The matrix identifies current public shelter, health, and service-continuity records as evidence gaps. No conclusion is made that facilities are absent or unable to operate; their current status is simply not verified in the public evidence base.

3.7 Recovery, livelihoods, and learning

The World Bank household data provides evidence relevant to poverty, housing conditions, flood exposure, and coping capacity in Addis Ababa and Dire Dawa (World Bank, 2017). This supports vulnerability and coping analysis, but it does not provide a current event-level recovery assessment.

The data describe 2017 conditions and are not exclusively Dire Dawa-specific. Public damage assessments, local disaster-office records, recovery-support records, and lessons-learned documents were not identified as usable current evidence. Evidence strength is therefore contextual for recovery planning.

It remains unknown how households currently recover, which livelihoods are most affected, how assistance reaches vulnerable groups, how services resume, and whether agencies systematically record and apply lessons from floods.

3.8 Cross-cutting evidence-quality and data gaps

Six included sources were scored in the internal quality screen. The resulting averages ranged from 2.67 to 3.67. No source reached the 4.0 threshold for strong operational evidence under the audit's screening framework. The city diagnostic and local survey evidence are useful but require qualification; the household vulnerability dataset is contextual or historical because of access and specificity limits.

Across domains, the most consistent limitations are age, incomplete geographic coverage, heterogeneous methods, restricted or uncertain reusability, and the absence of current end-to-end operational records. The evidence matrix contains 26 findings, including documented, partial, historical, and evidence-gap findings. It does not support a claim that Dire Dawa's complete warning chain is currently functional or non-functional.

The priority gaps are current warning reach and lead time, message comprehension and accessibility, route and shelter status, emergency-health continuity, responsible actors, and recovery learning.

Figure 1. Preparedness-chain evidence status. Source: evidence matrix; status reflects evidence strength and current-verification limits.

Figure 2. Latest evidence year represented by domain. Source: evidence matrix; a recent evidence year does not prove current operation.

4. Discussion

The central finding is that Dire Dawa has documented preparedness activity, but publicly accessible evidence is fragmented and insufficient to confirm current end-to-end operational readiness. The sources demonstrate that community warning approaches, formal warning arrangements, household preparedness research, and technical forecasting work have existed. They do not establish that all links remain functional, maintained, inclusive, or connected to safe action today.

Early warning must be assessed as a chain rather than as one tool. A siren is only one dissemination mechanism; it does not prove that a warning was generated from a reliable trigger, reached households, was understood, or led to feasible evacuation and shelter. Similarly, a technical model demonstrates analytical capacity but not an operational public alert service. The chain framing makes it possible to identify where evidence is strong enough to support a planning question and where current validation is necessary before implementation.

Data gaps matter for inclusive planning. People do not experience warnings as an abstract citywide service. Women may face different communication, care, and mobility constraints. Older persons and persons with disabilities may need accessible channels, assistance, or transport. People with limited phone access may depend on local networks, sirens, or trusted intermediaries. Informal settlers and people whose livelihoods cannot easily pause may face different costs when a warning is late or ambiguous. The reviewed public evidence does not provide enough current disaggregated information to determine whether these groups are reached equitably.

Old information remains useful as context. The historical case study shows that locally adapted warning channels were possible and that community structures were used. The qualitative study identifies questions about evacuation, shelters, and health links that remain relevant to validation. The household survey explains why threat perception, barriers, benefits, and cues to action matter. But none of these should be treated as present-day proof without current confirmation.

The study adds an evidence-quality and decision-readiness approach to existing flood-hazard work. Instead of asking only where flooding may occur or whether a model can be built, it asks what public evidence supports a safe decision, what remains uncertain, and what must be checked locally. This approach connects research to action while preserving the distinction between documented practice and current operational performance.

The limitations are material. The review relies on public and lawfully accessible sources, and relevant government, NGO, monitoring, health, shelter, and coordination records may exist outside the public evidence base. Source periods and geographies differ. The scorecard is an internal screening framework, not a validated international measure. No primary interviews, household survey, site inspection, drill, or institutional verification was undertaken. The conclusions should therefore guide validation and cautious planning, not substitute for it.

Figure 3. Evidence-quality screening by domain. Source: internal scorecard; values are not operational readiness scores.

5. Recommendations

Immediate no-regret actions

The following actions improve readiness for better decisions without assuming that unverified capabilities are absent or effective:

- Establish a common evidence inventory shared by relevant institutions, with source owner, date, geography, status, access level, and update frequency.
- Create a standard tool for verifying warning channels, roles, equipment, backup systems, language accessibility, disability access, and household reach.
- Define minimum indicators for warning performance, evacuation, shelter, emergency-health response, service continuity, and recovery.
- Create an inclusive field-validation protocol that protects personal information, avoids exposing vulnerable communities to harm, and records uncertainty.
- Establish a periodic public update process for non-sensitive preparedness data, including publication date, evidence period, responsible institution, and known limitations.

Actions requiring field validation first

- Expanding or replacing alert channels.
- Designating evacuation routes or assembly points.
- Naming shelters or estimating shelter capacity.
- Targeting specific kebeles for interventions.
- Making claims about who is or is not covered by warning systems.
- Publishing sensitive facility or location data.

These actions may be important, but public secondary evidence alone cannot establish that the selected channel, route, facility, or target population is appropriate today. The field-validation protocol should precede implementation and should include institutions and affected communities.

Figure 4. Research-to-action pathway. Source: analysis memo and evidence matrix.

6. Conclusion

Public evidence confirms that flood-preparedness and early-warning practices have been implemented in Dire Dawa, including community-based approaches and formal warning arrangements. However, the reviewed evidence is fragmented, often historical, and insufficiently standardized to verify current operational coverage, accessibility, evacuation readiness, shelter functionality, emergency-health continuity, or end-to-end warning performance.

The evidence supports a cautious interpretation. Dire Dawa has documented preparedness activity and technical and behavioural research, but no reviewed source provides strong operational proof across the complete warning chain. Historical evidence can guide questions and show that locally adapted approaches have been possible; it cannot by itself establish current function or equitable reach.

The priority is therefore not to assume absence or effectiveness, but to conduct responsible, location-specific field validation before program targeting and investment decisions are made. A common evidence inventory, standardized validation tools, minimum performance indicators, and a periodic public update process can make that next stage more transparent and useful.

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Appendices

Appendix A: Full source inventory

Complete field-level source inventory: 02_source_inventory.csv. The companion CSV contains URLs, methods, scores, limitations, and decision reasons.

Appendix B: Preparedness evidence matrix

Complete 16-column finding-level matrix: 03_preparedness_evidence_matrix.csv.

Appendix C: Evidence-quality rubric

The scorecard is a transparent internal screening framework developed for this evidence audit. It is not an internationally validated scale.

Calculation: Evidence Quality Score = (Recency + Geographic Specificity + Method Strength + Public Accessibility + Actionability + Reusability) / 6.

Interpretation: 4.0-5.0 = strong operational evidence; 3.0-3.9 = useful but requires qualification; 2.0-2.9 = contextual or historical evidence; below 2.0 = background only and not sufficient alone for recommendations. Years not explicitly bracketed in the rubric were treated conservatively by interpolation: 2012-2016 = 2 and 2021-2023 = 4.

Appendix D: Search terms and source-selection log

Search terms used in combinations included: Dire Dawa flood; Dire Dawa flash flood; Dire Dawa flood preparedness; Dire Dawa early warning; Dire Dawa flood forecasting; Dire Dawa evacuation; Dire Dawa shelter; Dire Dawa emergency health; Ethiopia urban flood risk; Dire Dawa vulnerability; Dire Dawa drainage; Dire Dawa recovery; community early warning Dire Dawa; and flood preparedness behaviour Dire Dawa.

Selection log: sources were retained when they provided direct Dire Dawa findings, a local study or case, a city-level diagnostic, technical analysis, or relevant vulnerability evidence. Sources were marked as missing or needing verification when the evidence type was decision-critical but the actual public document was not identified. General reports that only mentioned Dire Dawa briefly were excluded from primary evidence unless they added direct local or methodological value. Media and other triangulation sources were not treated as primary operational proof.

Appendix E: Data dictionary

Appendix F: Ethical and data-protection note

This report uses public or lawfully accessible secondary sources and does not collect personal data. It does not publish individual household information, identify sensitive facilities, map unverified evacuation routes, or rank communities by preparedness. Restricted datasets are described only at the level permitted by their public documentation. Where current evidence is missing, the report states that the capability is unverified or represents an evidence gap; it does not claim that the service is absent. Any future field validation should use informed participation, data minimization, secure handling, accessible communication, and a clear process for avoiding harm to vulnerable communities.

Publication tables

Domain	Findings	Status
Forecasting and warning generation	3	Documented, Evidence gap, Partial
Protective action and evacuation	4	Documented, Evidence gap, Historical, Partial
Recovery, livelihoods, and learning	4	Documented, Evidence gap
Risk monitoring and hazard information	3	Documented, Evidence gap, Partial

Domain	Findings	Status
Shelter, emergency health, and service continuity	4	Evidence gap, Partial
Warning dissemination	5	Evidence gap, Historical, Partial
Warning receipt, understanding, and accessibility	3	Evidence gap, Historical

Source: 03_preparedness_evidence_matrix.csv. The release package contains the full source inventory, matrix, rubric, and ethical note.